MATH O MAGIC

6·4

by

Greg Hurn

EDUCATIONAL DIRECTIONS PUBLICATIONS

INTRODUCTION

MATH O MAGIC 6·4 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

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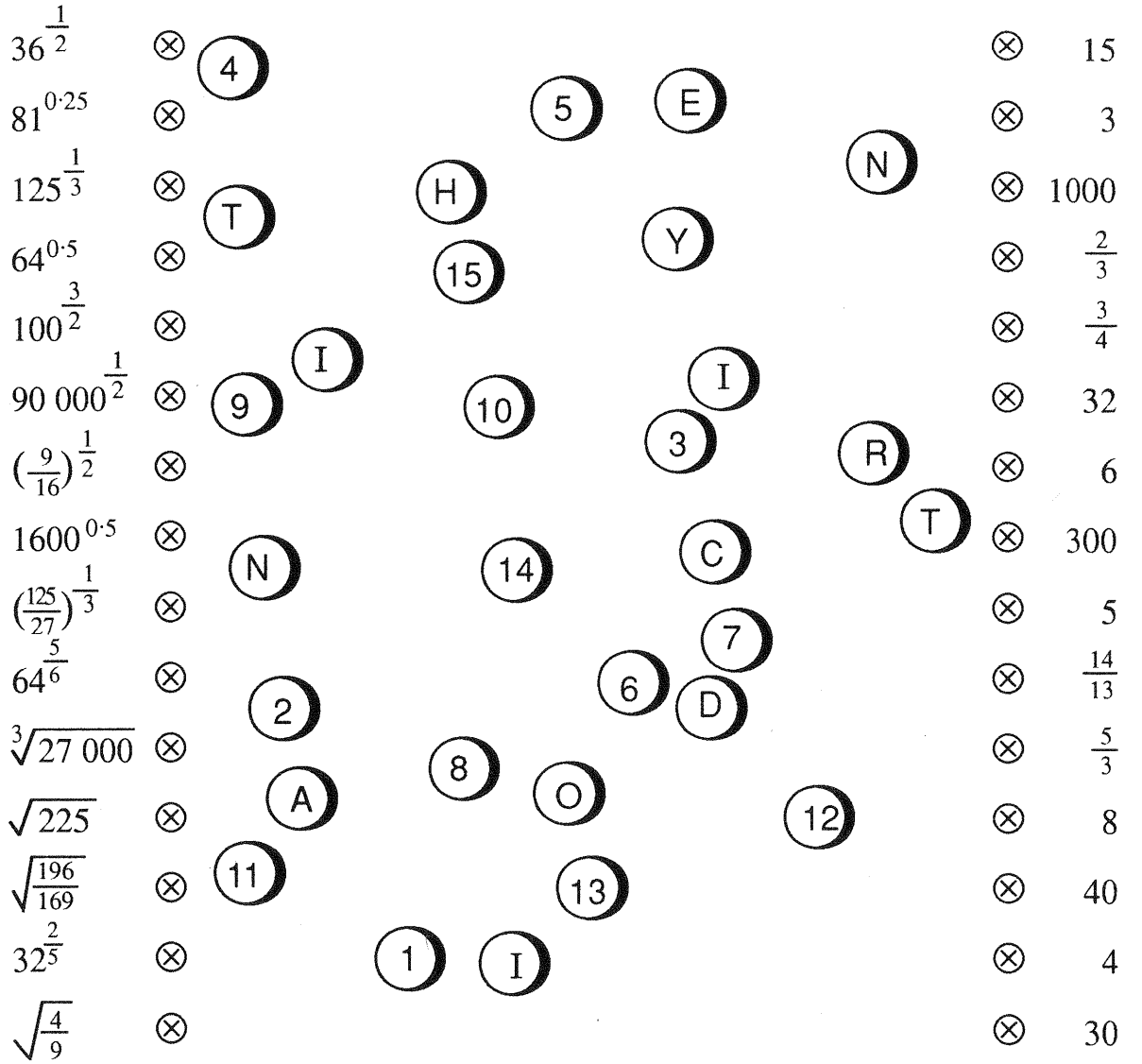
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WHERE DOES SUCCESS COME BEFORE WORK?

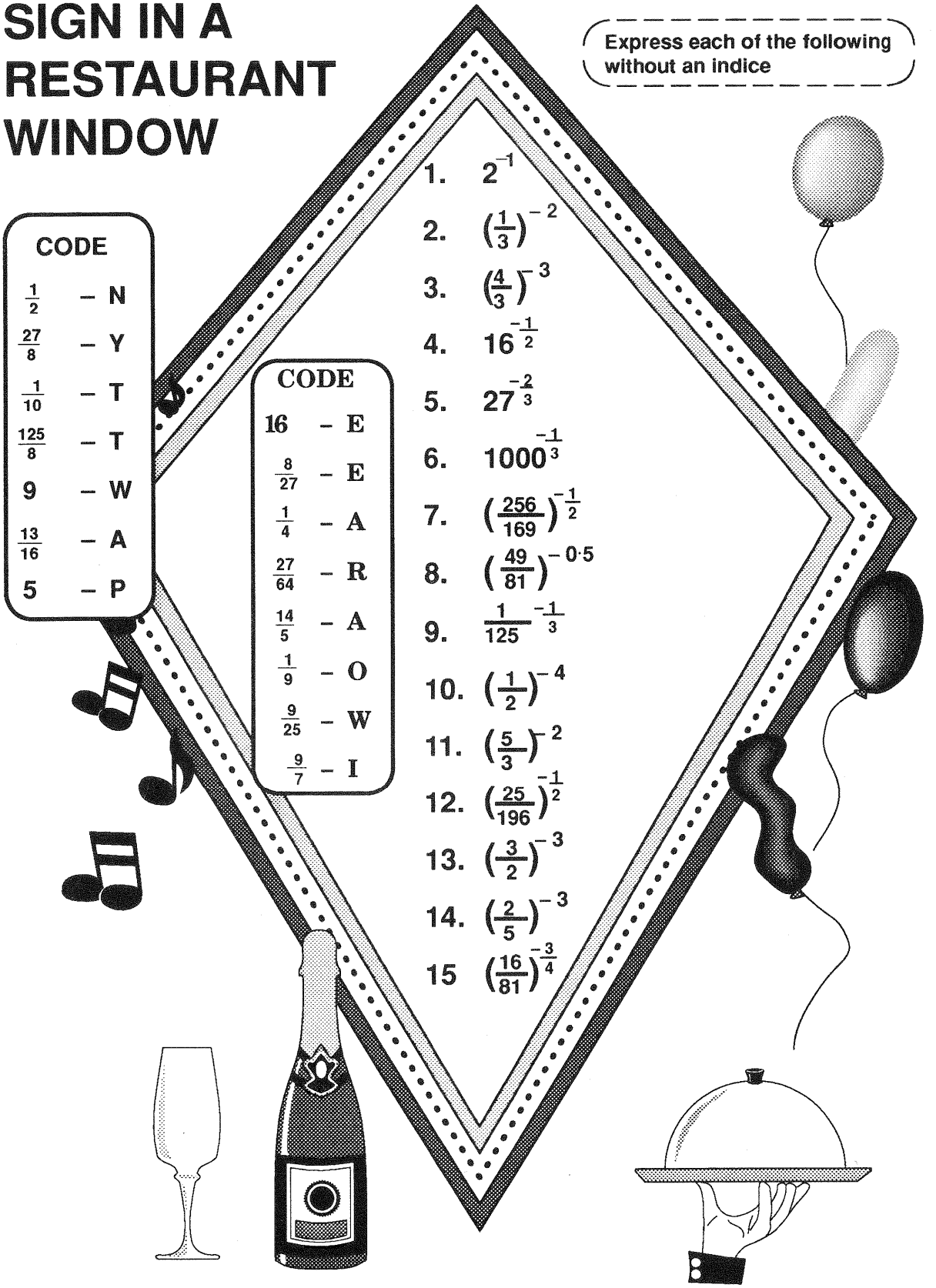
JOIN EACH OF THE EXPRESSIONS WITH THE EQUAL VALUE ON THE OTHER SIDE.



1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

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SIGN IN A RESTAURANT WINDOW



Express each of the following without an indice

1. 2^{-1}
2. $\left(\frac{1}{3}\right)^{-2}$
3. $\left(\frac{4}{3}\right)^{-3}$
4. $16^{-\frac{1}{2}}$
5. $27^{-\frac{2}{3}}$
6. $1000^{-\frac{1}{3}}$
7. $\left(\frac{256}{169}\right)^{-\frac{1}{2}}$
8. $\left(\frac{49}{81}\right)^{-0.5}$
9. $\frac{1}{125}^{-\frac{1}{3}}$
10. $\left(\frac{1}{2}\right)^{-4}$
11. $\left(\frac{5}{3}\right)^{-2}$
12. $\left(\frac{25}{196}\right)^{-\frac{1}{2}}$
13. $\left(\frac{3}{2}\right)^{-3}$
14. $\left(\frac{2}{5}\right)^{-3}$
15. $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$

CODE

$\frac{1}{2}$	N
$\frac{27}{8}$	Y
$\frac{1}{10}$	T
$\frac{125}{8}$	T
9	W
$\frac{13}{16}$	A
5	P

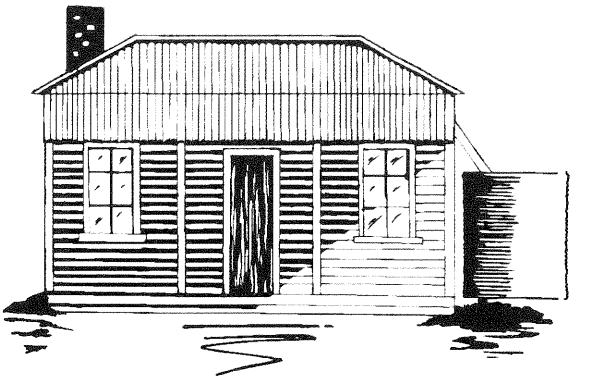
CODE

16	E
$\frac{8}{27}$	E
$\frac{1}{4}$	A
$\frac{27}{64}$	R
$\frac{14}{5}$	A
$\frac{1}{9}$	O
$\frac{9}{25}$	W
$\frac{9}{7}$	I

10	4	6	1	5	11	—	9	12	15	2	7	8	14	13	3

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HOW ARE COUNTRY PEOPLE UNUSUAL?



Find the appropriate values for each of the following expressions (to 1 d.p.)

1. $10^{0.4}$
2. $8^{2.1}$
3. $12^{0.7}$
4. $33^{1.9}$
5. $9^{3.1}$
6. $3^{5.7}$
7. $38^{0.28}$
8. $15^{\frac{3}{4}}$
9. $29^{\frac{2}{3}}$
10. $17^{-0.2}$
11. $2^{-1.3}$
12. $1.5^{8.8}$
13. $0.7^{-5.6}$
14. $0.9^{\frac{-5}{2}}$
15. $257^{-0.01}$
16. $30^{\frac{-1}{2}}$
17. $0.68^{-5.1}$

CODE

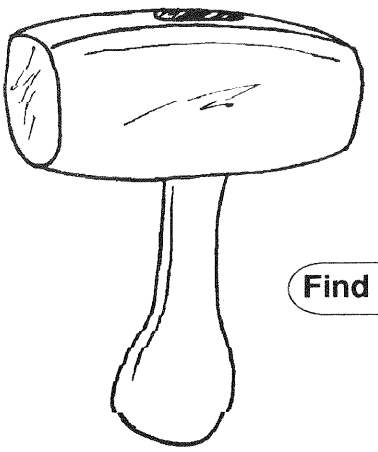
908·1	– E	2·8	– G
0·6	– S	0·9	– H
0·4	– I	78·8	– D
2·5	– T	7·1	– B
7·4	– O	5·7	– W
767·7	– H	1·3	– E
35·4	– C	9·4	– K
		0·2	– Y
		524·3	– O
		7·6	– T



1	4	5	16		7	13
8	6		17	14	2	
3	11	8	4		1	15	14
12	15	13	6	9	10	



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WHAT KIND OF TOOL DO YOU USE TO FIX A BROKEN GORILLA?

Find the value of x in each of the following equations?

E $2^x = 16$

O $8^x = 16$

A $27^x = 9$

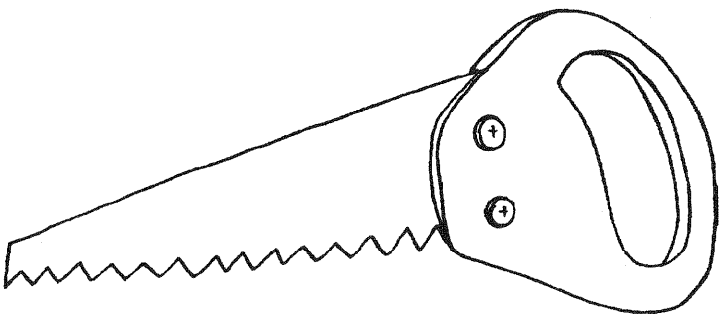
Y $64^x = \frac{1}{8}$

N $8^x = 128$

C $4^x = \frac{1}{32}$

N $8^{2x} = 16^{x+1}$

E $16^x = 128$



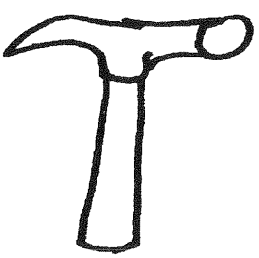
W $3^{3x} = 9^{x+3}$

H $8^{4-x} = 32^{2x}$

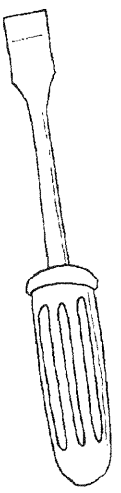
K $25^{3x-1} = 125^{x+3}$

M $4^{3-x} = 32^{2x+1}$

R $625^x = 125^{2x-1}$



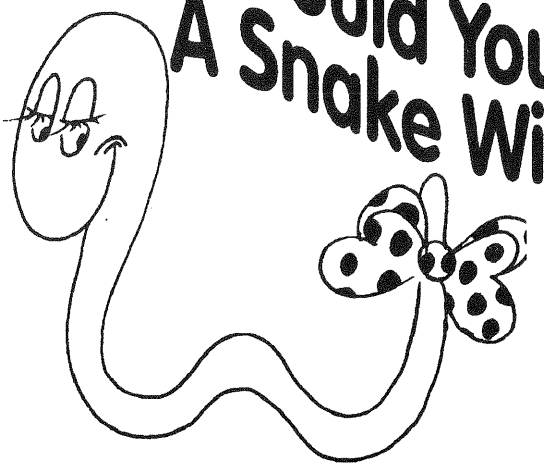




$\frac{2}{3}$	$\frac{1}{12}$	$\frac{4}{3}$	2	$\frac{11}{3}$	$\frac{7}{4}$	$\frac{-1}{2}$	6	$\frac{3}{2}$	4	$\frac{7}{3}$	$\frac{-5}{2}$	$\frac{12}{13}$

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What Would You Get if You Crossed A Snake With A Basketball?



Find the value of each of the following items at the end of the time indicated.

	INITIAL VALUE	INFLATION	DEPRECIATION	TIME (years)
U	$100	8%		3
E	$200	10%		4
A	$500		15%	2
C	$400		5%	5
O	$250	20%		3
I	$300		12%	6
Y	$420		7%	8
N	$615	9%		5
T	$374	7%		9
G	$298		13%	10
D	$437	5%		20
B	$248	12%		15

235·02	432	125·97	1159·49	74·03	292·82	687·58	361·25	1357·44	432	125·97	946·25	309·51	139·32	946·25	74·03

1357·44	361·25	1357·44	235·02	1357·44	432	361·25

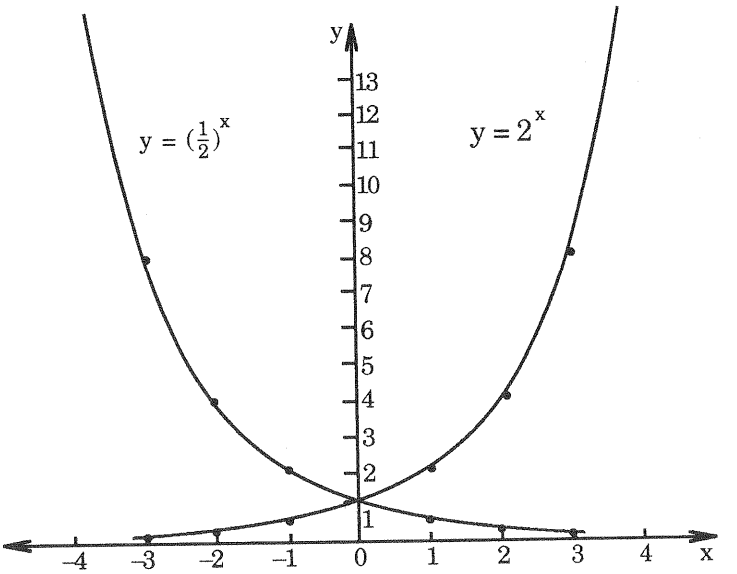
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WHAT DOCTOR DO YOU GO TO TO CURE A SICK EGYPTIAN MUMMY?



Use the following graphs to find a value for each of the variables.

A	2^x	$= 6$
C	2^x	$= 11$
I	2^x	$= 3$
O	$2^{0.5}$	$= y$
R	$2^{-1.3}$	$= y$
P	2^{2x}	$= 5$
A	2^{x+1}	$= 10$
R	2^{2x-1}	$= 6$
T	$(\frac{1}{2})^x$	$= 7$
O	$(\frac{1}{2})^x$	$= 2$
C	$(\frac{1}{2})^{-3}$	$= y$
R	$(\frac{1}{2})^{-1}$	$= y$
A	$(\frac{1}{2})^{1.8}$	$= y$



○	○	○	○	○	○	-	○	○	○	○	○	○	○
2·3	3·5	0·3	1·6	1·8	1·4		1·2	2	2·6	8	−2·8	−1	0·4

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How Did The Waiter Explain To The Customer About The Squashed Sandwich?

Factorize the following trinomials – Draw a line to the matching answer.

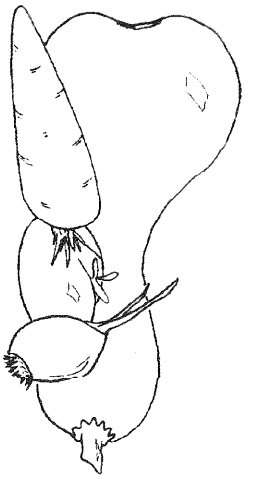
$2x^2 - 5x - 3$	$(x + 2)(x - 5)$
$4x^2 + 4x + 1$	$(2x + 1)(x - 3)$
$36x^2 - 11x - 12$	$(3x - 2)(x + 1)$
$12x^2 + 8x - 15$	$(4x - 1)(x + 3)$
$4x^2 + 11x - 3$	$(2x - 1)(x + 4)$
$21x^2 - 13x + 2$	$(2x + 1)(2x + 1)$
$x^2 - 3x - 10$	$(6x - 5)(2x + 3)$
$3x^2 + x - 2$	$(3x + 2)(2x + 5)$
$6x^2 + 19x + 10$	$(10x + 1)(2x - 5)$
$20x^2 - 48x - 5$	$(7x - 2)(3x - 1)$
$24x^2 + 22x - 35$	$(4x + 7)(6x - 5)$
$2x^2 + 7x - 4$	$(9x + 4)(4x - 3)$

9	T	S	7	M	6	2	8	L	12	10	Y	5	I	P	11	3	D	4	N
E	U	O	1																

2	1	5	9	1	8	3	7	4	9	1	6	9	4	12	1	11	10	9	

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WHAT DO YOU GET WHEN YOUR HORSE WALKS THROUGH YOUR VEGETABLE GARDEN?



Examine each of the equations below. Look at the value of each DISCRIMINANT to determine the number of solutions. Cross out the boxes with one or two solutions. The remaining letters in order will answer the riddle.

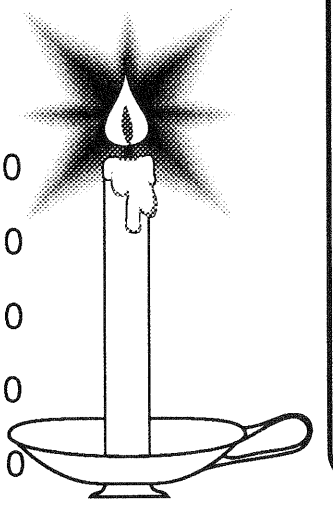
$x^2 + 2x + 3 = 0$ **S**	$x^2 - 4x + 2 = 0$ **M**	$x^2 + 5x + 1 = 0$ **R**
$x^2 + 2x - 3 = 0$ **E**	$x^2 + 3x + 2 = 0$ **T**	$3x^2 - 2x + 1 = 0$ **Q**
$2x^2 - 5x + 1 = 0$ **T**	$3x^2 + 4x - 2 = 0$ **D**	$5x^2 - 3x + 1 = 0$ **U**
$4x^2 - 6x + 2 = 0$ **O**	$x^2 + 6x + 9 = 0$ **C**	$4x^2 + 4x + 1 = 0$ **A**
$3x^2 - 2x + 5 = 0$ **A**	$4x^2 - 3x - 2 = 0$ **Y**	$8x^2 - 6x - 3 \neq 0$ **G**
$2x^2 + 7x + 2 = 0$ **V**	$5x^2 - 9x + 4 = 0$ **N**	$3x^2 + 7x + 9 = 0$ **S**
$6x^2 - 5x - 3 = 0$ **R**	$9x^2 + 24x + 16 = 0$ **D**	$10x^2 - 3x + 2 = 0$ **H**

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WHY IS AN EVIL PERSON LIKE A CANDLE?

Solve each of the following equations and find the solutions in the code key. Use the code to answer the riddle.

1. $x^2 - 7x + 12 = 0$
2. $x^2 - 3x - 10 = 0$
3. $x^2 - 10x + 25 = 0$
4. $x^2 - 36 = 0$
5. $x^2 - 81 = 0$
6. $x^2 + 7x = 0$
7. $6x^2 - 5x - 4 = 0$
8. $15x^2 - x - 2 = 0$
9. $4x^2 - 33x - 27 = 0$
10. $8x^2 - 26x + 21 = 0$
11. $6x^2 + 17x - 45 = 0$
12. $15x^2 - 32x - 7 = 0$
13. $16x^2 + 24x + 9 = 0$



CODE KEY

B	–	$\frac{2}{5}, \frac{-1}{3}$
W	–	$\frac{-3}{4}, \frac{-3}{4}$
E	–	3, 4
H	–	5, 5
K	–	$\frac{-9}{2}, \frac{5}{3}$
R	–	– 2, 5
C	–	$\frac{7}{3}, \frac{-1}{5}$
T	–	6, – 6
I	–	$\frac{3}{2}, \frac{7}{4}$
A	–	9, – 9
Y	–	– 7, 0
O	–	$9, \frac{-3}{4}$
D	–	$\frac{-1}{2}, \frac{4}{3}$

4	3	1	6	5	2	1	8	9	4	3	
13	10	12	11	—	1	7					

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WHAT'S THE DIFFERENCE BETWEEN A HUNGRY PERSON AND A GREEDY PERSON?

Solve each of the following equations.
Use the answers to solve the riddle.

A $\frac{5}{2a} = \frac{4}{9}$

N $\frac{11}{4} = \frac{5}{2e}$

O $\frac{2}{7} = \frac{3}{x}$

H $\frac{3}{3h+1} = 5$

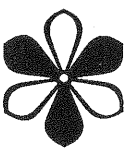
E $\frac{4}{3-e} = 6$

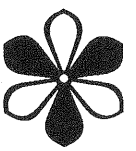
R $\frac{5}{2r-1} = 4$

T $\frac{4}{2-t} = \frac{3}{3t+1}$









$\frac{g+2}{4} = \frac{g}{3}$ **G**

$\frac{2}{3+s} = \frac{4}{2+s}$ **S**

$\frac{5}{2-2e} = \frac{3}{2e+1}$ **E**

$\frac{4}{x+3} = \frac{3}{x-1}$ **O**

$\frac{2t-1}{5} = \frac{t+2}{2}$ **T**

$\frac{p+1}{3} = 2 + \frac{3p-1}{4}$ **L**

10·5	$0\cdot\overline{90}$	$2\cdot\dot{3}$	− 3·4	13	$0\cdot\overline{90}$	6	− 4	$0\cdot1\dot{3}$	10·5	0·0625	5·625	− 12	$0\cdot1\dot{3}$	$-1\cdot\dot{3}$	$2\cdot\dot{3}$
13	− 12	$-1\cdot\dot{3}$	$2\cdot\dot{3}$	1·125	0·0625	5·625	− 12	− 4	$0\cdot1\dot{3}$	13	10·5	− 3·4	13	$0\cdot\overline{90}$	6

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WHY ARE MUMMIES SO NERVOUS?

Use the quadratic formula to find solutions to the following equations. All answers occur in the sentence at the bottom of the page (some occur twice).

A $8x^2 - 2x - 3 = 0$

E $10x^2 + x - 3 = 0$

U $4x^2 + 9x + 2 = 0$

N $9x^2 + 6x + 1 = 0$

L $4x^2 - 21x + 20 = 0$

T $x^2 + 6x + 9 = 0$

O $4x^2 - 20x + 25 = 0$

S $x^2 - 10x + 25 = 0$

H $25x^2 - 40x + 16 = 0$

E $10x^2 - 39x + 14 = 0$

Y $x^2 + 8x + 16 = 0$

B $4x^2 - 12x + 9 = 0$

C $4x^2 - 36x + 81 = 0$

D $x^2 + 20x + 100 = 0$

P $x^2 - 12x + 36 = 0$

R $25x^2 - 60x + 36 = 0$

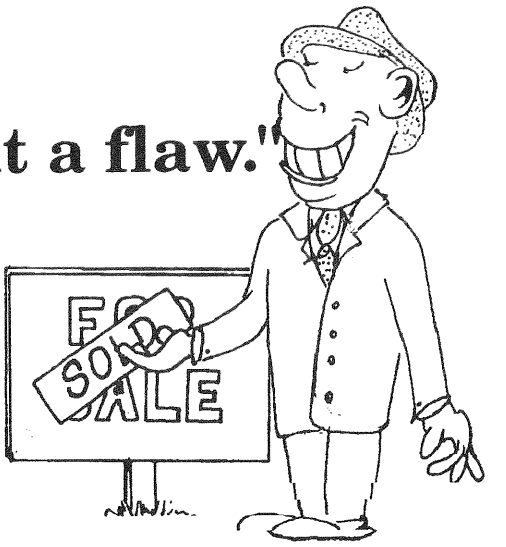
W $16x^2 - 56x + 49 = 0$

Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ
1·5		·4	4·5		−·5	−2	5		·5	−3		·8	−·6	−4		·75	1·2		3·5		

Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ	Δ
−·5		4	1·25		1·75		2·5		−·25		−·33		−10		−2		6				

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REAL ESTATE AGENT:
"Here is a house without a flaw."



Find the approximate solution to each of these quadratic equations, using the quadratic formula.

Y $x^2 + 3x - 2 = 0$

O $x^2 + 2x - 5 = 0$

A $x^2 + 5x + 2 = 0$

S $x^2 - 7x + 4 = 0$

D $x^2 + 4x + 2 = 0$

W $x^2 - 3x - 4 = 0$

L $x^2 - x + 0{\cdot}25 = 0$

H $x^2 - 3x + 2{\cdot}25 = 0$

T $x^2 + 6x + 9 = 0$

M $x^2 - 12x + 36 = 0$

U $x^2 - 5x + 6{\cdot}25 = 0$

K $x^2 + 3x + 2{\cdot}25 = 0$

E $x^2 - 7x + 12{\cdot}25 = 0$

O $x^2 + 7x - 6 = 0$

N $x^2 - 3x - 5 = 0$

G $x^2 + 5x + 6{\cdot}25 = 0$

CUSTOMER:

"															
6	0·56		−2·5	1·45	0·77	−0·59	−1·19	3·5	6·37	0·63	,		−1	1·5	−4·56	−3
															"	
	−3·41	−3·45		−3·56	0·77	2·5		4	−0·44	0·5	−1·5			−7·77	4·19	

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The Boy Was Late For School Because Of A Sign – WHAT DID IT SAY?

Using the quadratic formula, find the approximate solution for each of the following equations.



O $2x^2 + 3x - 2 = 0$

H $3x^2 + 2x - 3 = 0$

L $5x^2 - 3x - 1 = 0$

D $9x^2 - 6x + 1 = 0$

A $4x^2 - 7x + 2 = 0$

E $16x^2 - 24x + 9 = 0$

C $16x^2 + 8x + 1 = 0$

W $4x^2 - 12x + 9 = 0$

O $2x^2 - 10x + 7 = 0$

S $3x^2 - 9x + 2 = 0$

G $16x^2 - 40x + 25 = 0$

2·76	−0·25	0·72	4·16	−2	0·84	1·39	−1·39	0·75	0·36	0·33

I						
	1·25	0·5	0·24	−0·24	0·84	1·5



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What Is A Midget Skunk Called?

$[\![\ ]\!]$ means the greatest integer less than or equal to a particular number.

Find the value of each of the following expressions.

A $[\![2{\cdot}7]\!]$

U $[\![-\frac{7}{4}]\!]$

K $[\![-0{\cdot}48]\!]$

N $[\![-4{\cdot}5]\!]$

K $[\![-6]\!]$

N $[\![-3\frac{1}{2}]\!]$

S $[\![3{\cdot}3]\!]$

U $[\![-7{\cdot}8-2]\!]$

K $[\![-2{\cdot}2 \times 4]\!]$

S $[\![-7\frac{3}{4}+1]\!]$

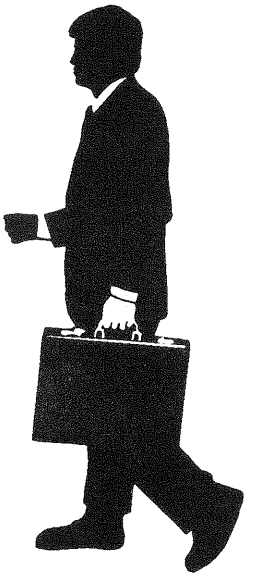
H $[\![-4{\cdot}9]\!]+9$

R $[\![2{\cdot}9 \times -3]\!]+9$

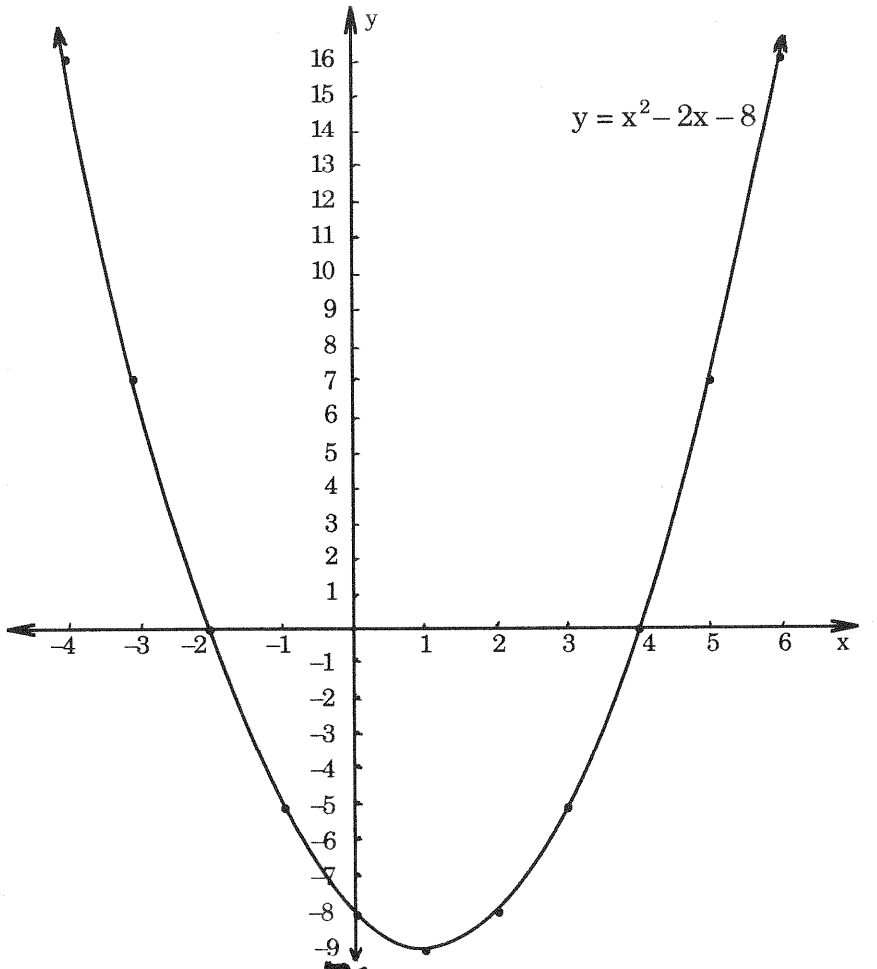
	2		−7	4	0	−10	−5	−1	3	−9	−2	−4	−6

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WHY DID THE BANK CLERK CLIMB A TREE?



Draw the graphs of suitable lines on the graph of the parabola below and use them to find the approximate solutions to the equations.





A $x^2 - 2x - 8 = 0$

B $x^2 - 2x - 8 = -5$

C $x^2 - 2x - 8 = 7$

O $x^2 - 2x - 8 = 16$

N $x^2 - 2x - 8 = 3$

E $x^2 - 2x - 5 = 0$

M $x^2 - 2x - 19 = 0$

R $x^2 - 2x + 0{\cdot}8 = 0$

A $x^2 - 2x - 22 = 0$

T $x^2 - 2x + 1 = 0$

H $x^2 - 2x - 8 = -10$

G $x^2 - 2x - 1{\cdot}3 = 0$

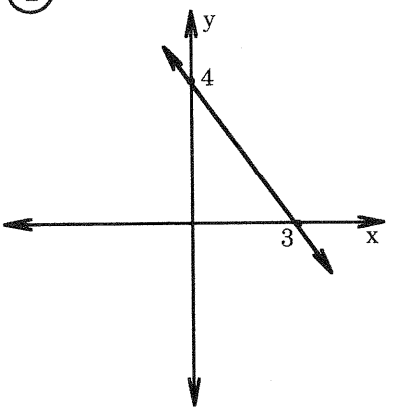
1	6		- 1	3·5	- 3	- 4	- 3·5	- 1·5		5·75

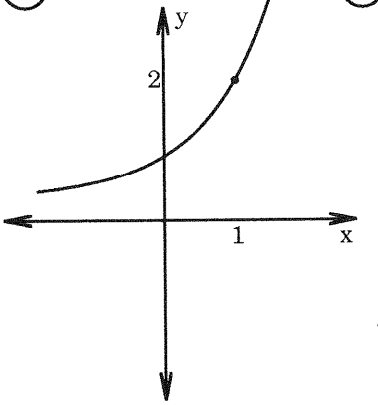
3	1·5	4	4·5	5	N.S.		5·5	- 2	- 2·5	- 3·75	2·5	3·5	0·5

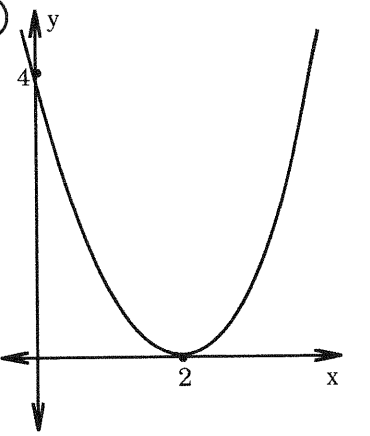
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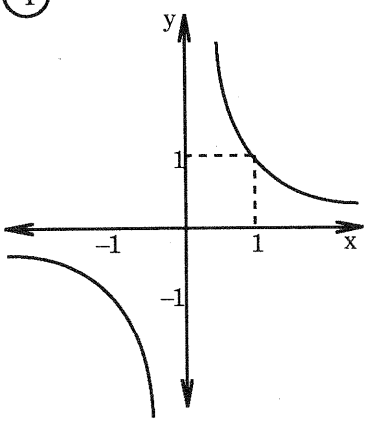
BILL : "A Man Sold Me The Nile River."

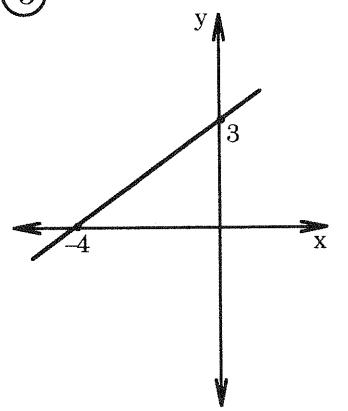
Identify the labels from the code key for each of the following graphs.

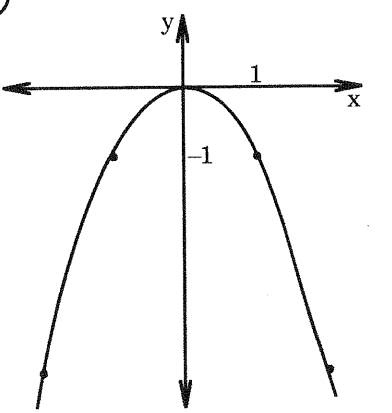
① 

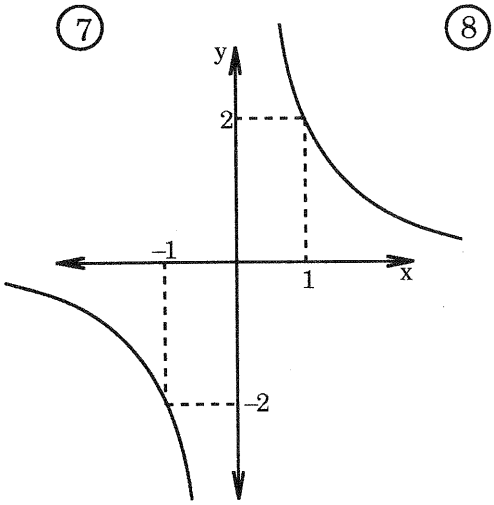
② 

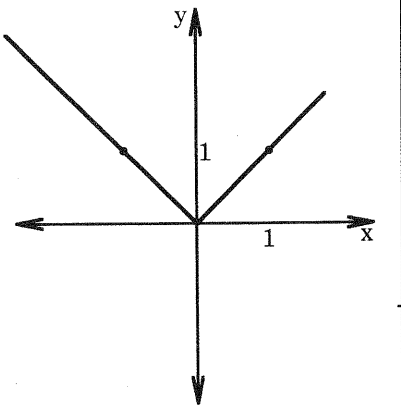
③ 

④ 

⑤ 

⑥ 

⑦ 

⑧ 

CODE KEY	
$y = -x^2$	Y
$y = \lvert x \rvert$	U
$y = 2^x$	G
$y = (x-2)^2$	Y
$xy = 1$	P
$y = \frac{2}{x}$	O
$3x - 4y = -12$	T
$4x + 3y = 12$	E

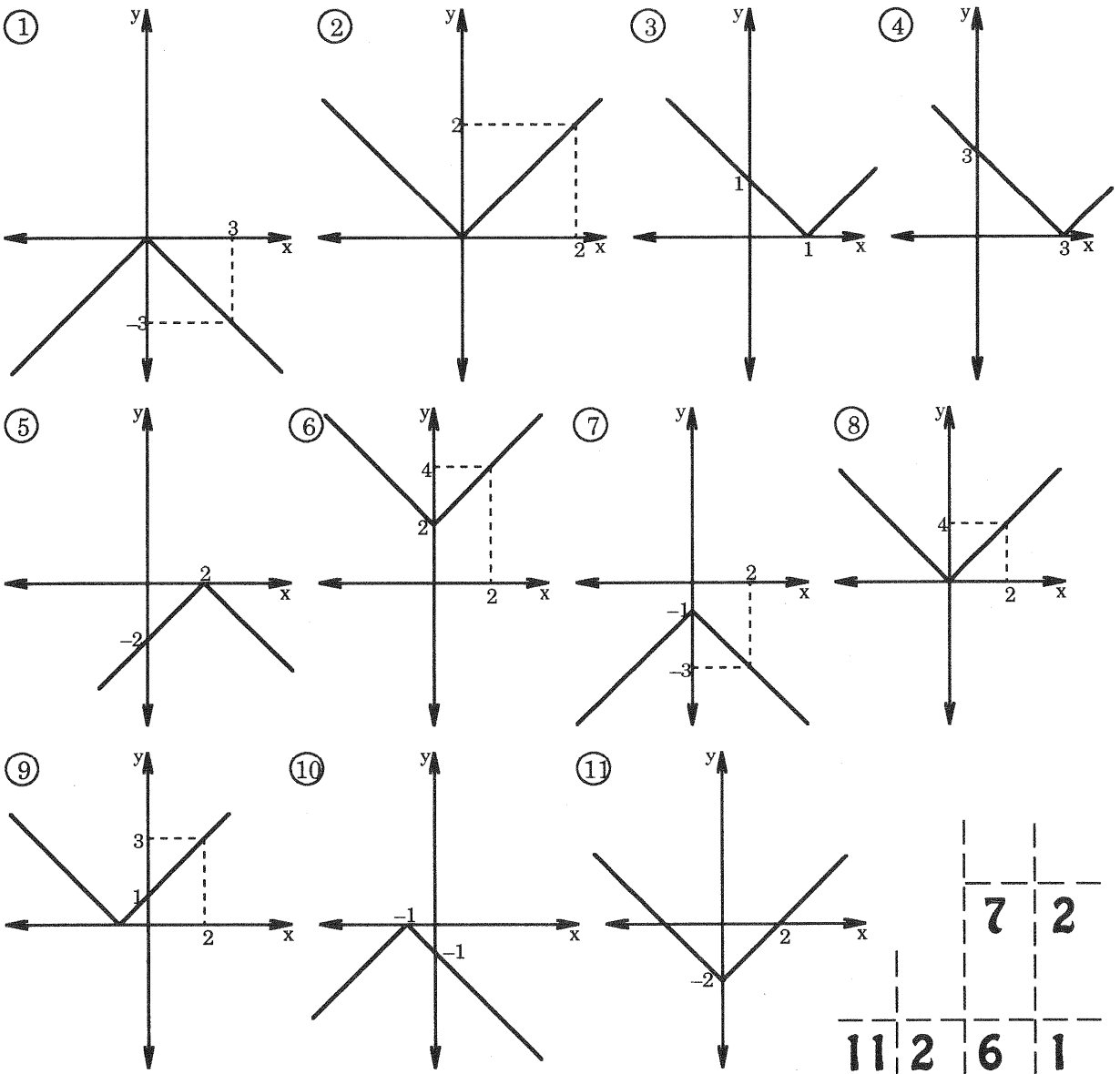
FRIEND :

"								"
1	*2*	*3*	*4*	*5*		*6*	*7*	*8*

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WHAT DID THE BOY MEAN WHEN HE TOLD HIS FATHER THAT HE DID THE SAME AS GEORGE WASHINGTON?

Identify each of these graphs from the code key.



CODE KEY			
$y = -\|x-2\|$	O	$y = \|2x\|$	Y
$y = -\|x\|-1$	H	$y = -\|x+1\|$	D
$y = \|x+1\|$	S	$y = \|x\| + 2$	N
$y = \|x\|$	E	$y = \|x\| - 2$	W
$y = \|x-1\|$	I	$y = -\|x\|$	T
		$y = \|x-3\|$	R

			7	2		
11	2	6	1			
10	5	11	6			
			3	6		
7	3	9	1	5	4	8

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Find the answers to each of the following questions at the bottom of the page.

WHY DID THE MAD SCIENTIST INVENT A SQUARE BATHTUB?

F 150 toys cost $2250. How much would 130 similar toys cost?

O A cyclist travels 840 m in 21 seconds. How far would he cycle in 17 seconds?

S A bag of 75 lollies weighs 450 g. How much would 92 similar lollies weigh?

T 9 cricket balls cost $108. How much would 17 cricket balls cost?

V A cake mix requires 35 g of sugar with 525 g of flour. How much flour is needed with 60 g of sugar?

L If a drive of 595 km uses 85 L of petrol, how far would 108 L go?

N A machine produces 2244 products in 17 minutes. How many would it produce in 24 minutes?

A If 13 people pack 3822 cartons in one day, how many cartons would 8 people pack?

G If 4 shop assistants serve 140 people during a lunch hour, how many people would be needed to serve 245 people?

I 74 m of timber flooring costs $333. How many metres could you buy for $432?

E A mowing contractor mows 384 m^2 in 16 minutes. How long would it take to mow 888m^2 ?

R A redwood cedar grows an average of 884 cm in 17 years. How long would it take to grow 1092 cm?

○	○		○	○		○	○	○	○	○		○	○	○	○				
552	680		96	204		3168	37	900	37	21		756	37	1950	204				
						○		○	○	○	○								
						2352		21	96	3168	7								

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WHY WAS THE MARINE CORPORAL KICKED OUT?

The answer to each question occurs in the sentence at the bottom of the page.

A If 8 people take 6 days to plant a rice paddy, how long would it take 12 people?

R A haystack can feed 15 cows for 8 days. How long would it last for 20 cows?

O A plane takes 12 hours at 750 km/hr to complete a flight. How long would it take at 1000 km/hr?

N If you could afford to buy 72 items at $4·50 each, how many could you buy if the price rose to $6·48?

W A life raft has enough supplies to support 10 people for 16 days. How long could they last 8 people?

H A cook decides that she has enough cereal for the 60 students for 5 days. If 40 of the students left, how many breakfasts would the cereal last for?

S A factory which employs 7 people has enough work for 20 days. However, if the work needs to be done in 14 days, how many MORE people are needed?

E A container of birdseed would feed a boy's 3 canaries for 50 days. If he buys another 2 birds, how long will the seed last?

C A manufacturer was going to make 84 coathangers, each using 80 cm of wire. He then decided to make a children's size using 60 cm of wire. How many could he make?

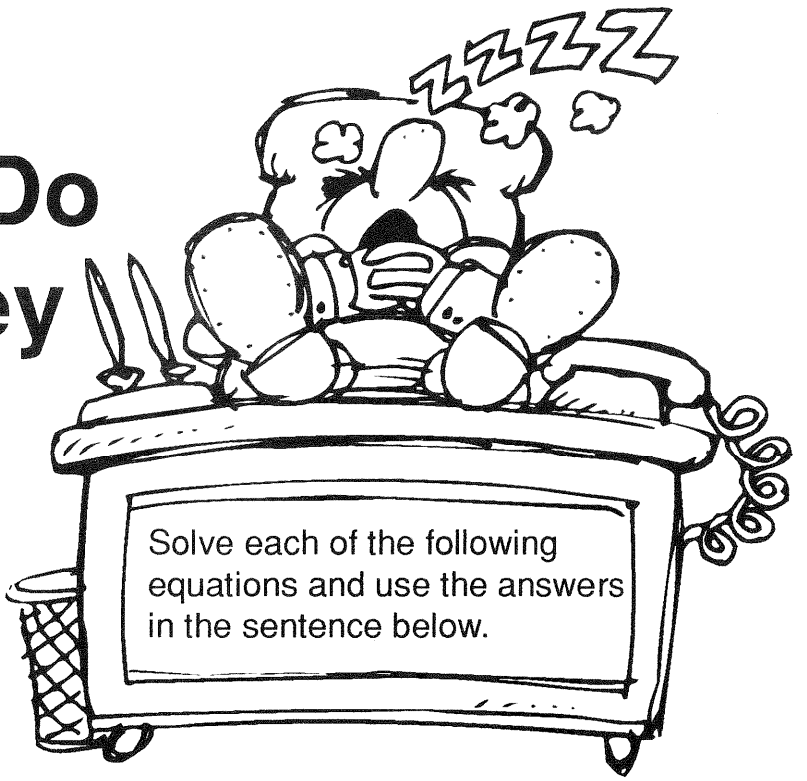
T A team of 8 people cleaning rubbish from the roadside could cover a stretch of road in 24 days. If two people left the group, how long would it take to cover the same distance?

P It took 10 window cleaners 21 days to wash all the windows in a high rise building. If 5 more cleaners were employed, how long would it take?

15	30	20	4	3	6	9	32	32	30	50	32	9	32	15	30	112	9	6	14	3

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What Do Termites Do When They Want To Relax?



$2^{t} = 16$

$2^{f} = 32$

$4^{a} = 16$

$16^{e} = 2$

$4^{k} = 32$

$27^{o} = 3$

$25^{a} = 5$

$2^{l} = \frac{1}{2}$

$9^{b} = \frac{1}{3}$

$27^{e} = 9$

$125 = 25^{b}$

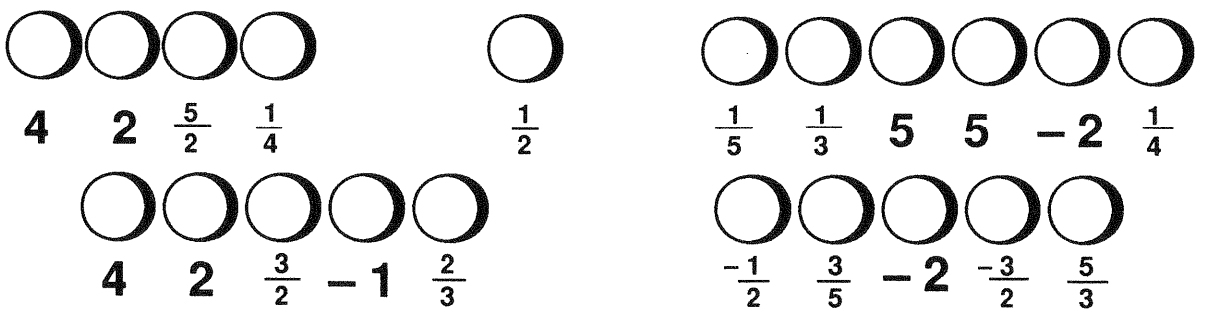
$32^{r} = 8$

$243 = 27^{k}$

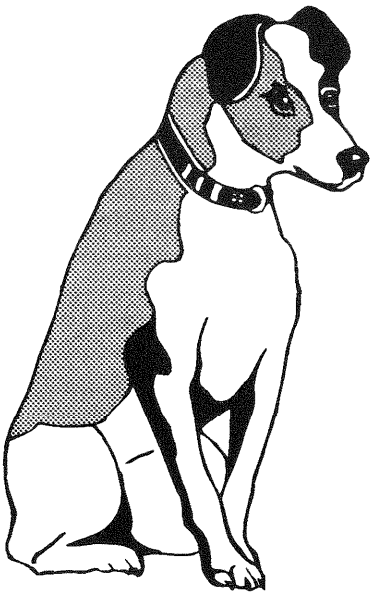
$9^{a} = \frac{1}{27}$

$32^{c} = 2$

$5^{e} = \frac{1}{25}$



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WHAT DO YOU NEED TO KNOW TO TEACH A DOG TRICKS?

Use index laws to solve the following equations.

$64^{e} = \frac{1}{8}$

$81^{a} = 27$

$4^{h} = \frac{1}{32}$

$125^{o} = 25$

$8^{t} = \frac{1}{128}$

$32 = 16^{n}$

$81 = 27^{e}$

$\frac{1}{125} = 625^{t}$

$8^{h+1} = 16^{h}$

$25^{r-1} = 125^{r}$

$16^{3d} = 8^{d+3}$

$3^{m+1} = 9^{2m-2}$

$4^{3g} = 8^{g+3}$

$4^{2-o} = 32^{o}$

□ □ □ □

$\frac{5}{3}$ $\frac{2}{3}$ -2 $\frac{-1}{2}$

□ □ □ □

$\frac{-3}{4}$ 3 $\frac{3}{4}$ $\frac{5}{4}$

□ □ □

$\frac{-7}{3}$ $\frac{-5}{2}$ $\frac{4}{3}$

□ □ □

1 $\frac{4}{7}$ 3

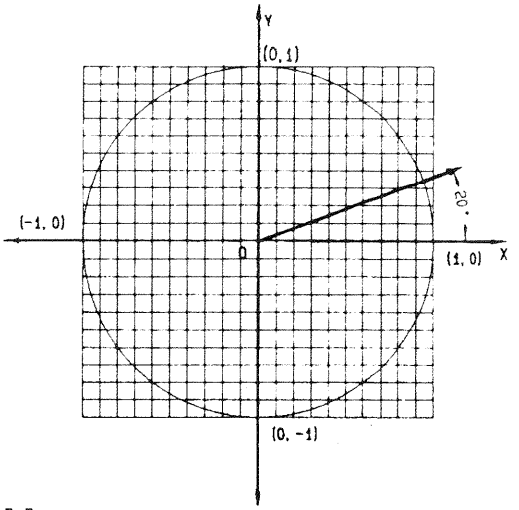


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WHY DID THE COFFEE TASTE LIKE MUD?

Use the **unit circle** to find a value for each of the following.



A sin 30°

I cos 45°

O sin 65°

H cos 120°

E sin 125°

S cos 127°

R sin 168°

N sin 17°

T cos 154°

U sin 24°

Y sin 90°

W cos 102°

R sin 143°

D cos 90°

S cos 114°

L cos 84°

F cos 107°

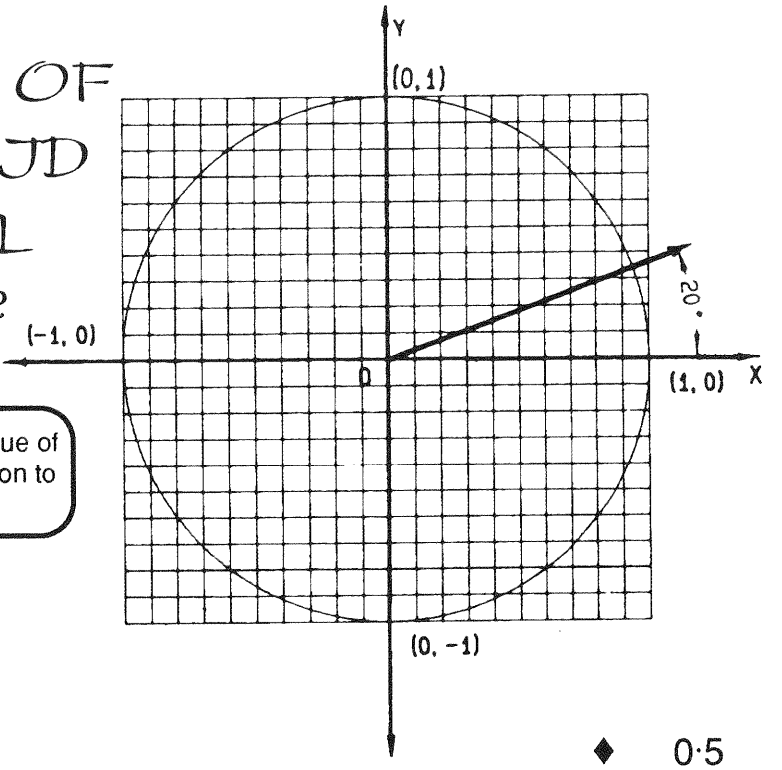
G cos 96°



0·7	−0·9		−0·2	0·5	−0·6	
−0·3	0·2	0·8	−0·4	−0·5	0·1	1
−0·1	0·6	0·9	0·4	0·3	0	

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WHAT SORT OF WORKERS DJD THE HOTEL REQUJRE?



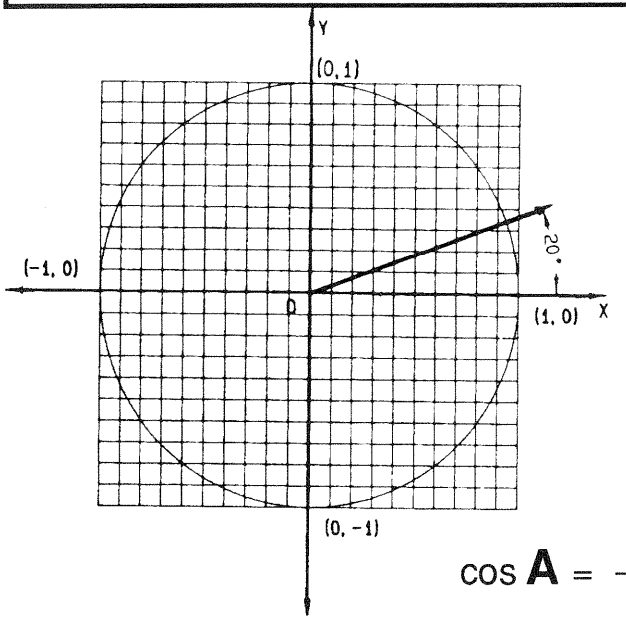
Use the UNIT CIRCLE to find the value of each of the following. Join the question to the answer with a straight line.

sin 270° ♦	N	♦ 0·5
sin 204° ♦	4 E	♦ – 0·4
cos 240° ♦	11 13 C D 14	♦ 0
sin 180° ♦	E 12	♦ 0·3
cos 300° ♦		♦ – 0·5
sin 296° ♦	10	♦ – 0·8
sin 217° ♦	E 6	♦ – 1
cos 282° ♦	I 1	♦ 0·2
cos 288° ♦	8	♦ – 0·7
sin 197° ♦	N R 5	♦ – 0·6
cos 225° ♦	3 X	♦ – 0·3
sin 307° ♦	P E 7	♦ 0·8
cos 360° ♦	2 N	♦ 1
cos 323° ♦	9 I	♦ – 0·9

1	2	3	–	4	5	6	7	8	9	10	11	12	13	14

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WHY WAS THE MAN SURE HE'D FIND HIS LOST DOLAR QUICKLY?



Use the UNIT CIRCLE to find all values (most have 2) for each angle, between 0° and 360°. Some of the angles occur in the sentence at the bottom of the page.



$\cos A = -0{\cdot}8$

$\sin D = -0{\cdot}5$

$\sin O = -1$

$\cos N = -1$

$\sin A = 0{\cdot}2$

$\cos E = 0{\cdot}3$

$\cos S = -0{\cdot}5$

$\sin R = -0{\cdot}7$

$\sin O = 0{\cdot}8$

$\sin T = -0{\cdot}1$

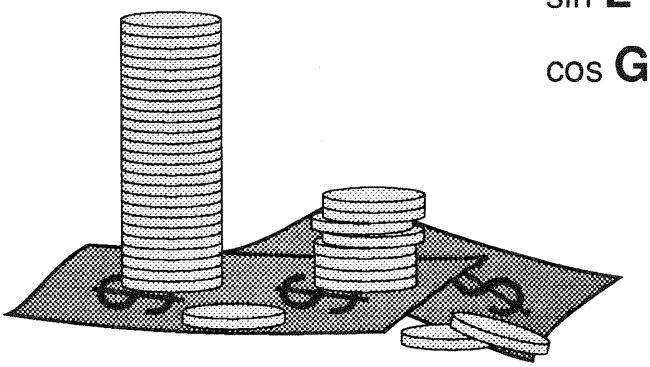
$\cos Y = 1$

$\cos F = -0{\cdot}2$

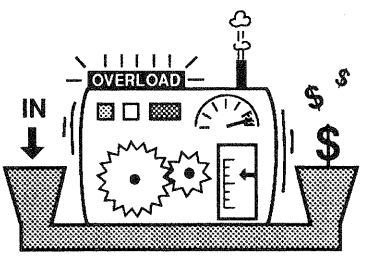
$\sin H = -0{\cdot}9$

$\sin L = -0{\cdot}4$

$\cos G = -0{\cdot}9$



12°	210°	127°	204°	336°	168°	315°	210°	53°	73°	120°	180°	354°



						154°	270°	102°	143°	225°			
186°	244°	287°	240°	287°				330°	216°	360°	240°		

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SIGN ON NEWLY SEEDED LAWN

Find a value for the variable in each of the equations below.
All angles are between 0° and 90°.

O $2 \sin x = 1{\cdot}6$

C $\cos x + 1 = 1{\cdot}4$

A $3 \tan x - 1 = 2{\cdot}7$

S $4 \sin x + 2 = 4{\cdot}2$

N $2 \cos x - 3 = -1{\cdot}58$

I $4 + 3 \sin x = 4{\cdot}93$

U $5 \tan x + 4 = 6{\cdot}3$

G $2 \sin x + 5 = 5{\cdot}28$

M $3(\cos x - 2) = -3{\cdot}38$

E $\frac{1}{2}(1 + \sin x) = 0{\cdot}86$

W $0{\cdot}3 \cos x - 3{\cdot}9 = -3{\cdot}87$

R $\frac{3}{4} \tan x + 2{\cdot}7 = 5{\cdot}95$

V $\frac{6 \sin x - 3}{2} = 1{\cdot}34$

D $4 - 3 \cos x = 1{\cdot}08$

B $4{\cdot}3 \tan x + 1{\cdot}9 = 5{\cdot}38$





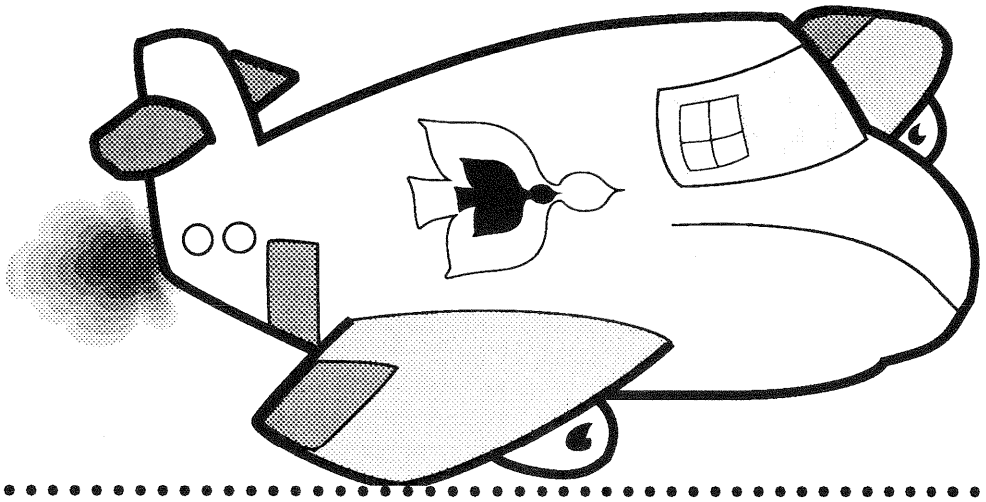
13° 53° 8° 33° 39° 46° 84° 51° 77° 46°



71° 18° 66° 18° 53° 25° 33° 29° 51° 45°



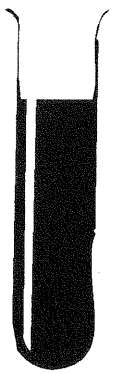
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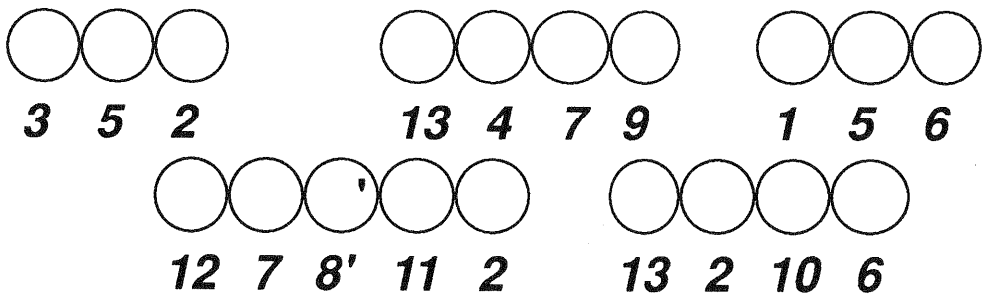
HOW IS AVIATION LIKE A DEADLY POISON?

Use your calculator to find two angles between 0° and 360° that satisfy these equations.

1. $\sin x = 0{\cdot}38$
2. $\cos x = -0{\cdot}57$
3. $\tan x = 1{\cdot}83$
4. $\tan x = -0{\cdot}92$
5. $\sin x = -0{\cdot}52$
6. $\sin x = -0{\cdot}26$
7. $\cos x = -0{\cdot}78$
8. $\tan x = 2{\cdot}08$
9. $\tan x = -0{\cdot}57$
10. $\cos x = 0{\cdot}21$
11. $\sin x = 0{\cdot}98$
12. $\sin x = -0{\cdot}44$
13. $\cos x = -0{\cdot}72$



CODE KEY	
U	64°, 244°
A	78°, 282°
A	22°, 158°
N	211°, 329°
O	141°, 218°
Y	206°, 334°
O	61°, 241°
D	195°, 345°
P	150°, 330°
R	79°, 101°
E	125°, 235°
D	136°, 224°
R	137°, 317°



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MATH O MAGIC 6·4 SOLUTIONS

1. IN THE DICTIONARY
2. EAT NOW – PAY WAITER
3. THEY GO TO BED WITH THE CHOOKS
4. A MONKEY WRENCH
5. YOU'D GET A BOUNCING BABY BOA
6. A CAIRO – PRACTOR
7. YOU TOLD ME TO STEP ON IT
8. SQUASH
9. THEY ARE BOTH WICK – ED
10. ONE LONGS TO EAT, THE OTHER EATS TOO LONG
11. BECAUSE THEY ARE ALL WOUND UP
12. MY GOODNESS, WHAT DO YOU WALK ON?
13. SCHOOL AHEAD – GO SLOW
14. A SHRUNK SKUNK
15. TO BECOME A BRANCH MANAGER
16. EGYPT YOU
17. HE WENT DOWN IN HISTORY
18. SO IT NEVER LEFT A RING
19. HE WAS ROTTEN TO THE CORPS
20. TAKE A COFFEE TABLE BREAK
21. MORE THAN THE DOG
22. IT WAS FRESHLY GROUND
23. INN – EXPERIENCED
24. A DOLLAR DOESN'T GO FAR THESE DAYS
25. DOGS BEWARE, VICIOUS MAN
26. ONE DROP AND YOU'RE DEAD

